Week 8: Student Performance Dashboard

(Power BI Desktop)

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Aim

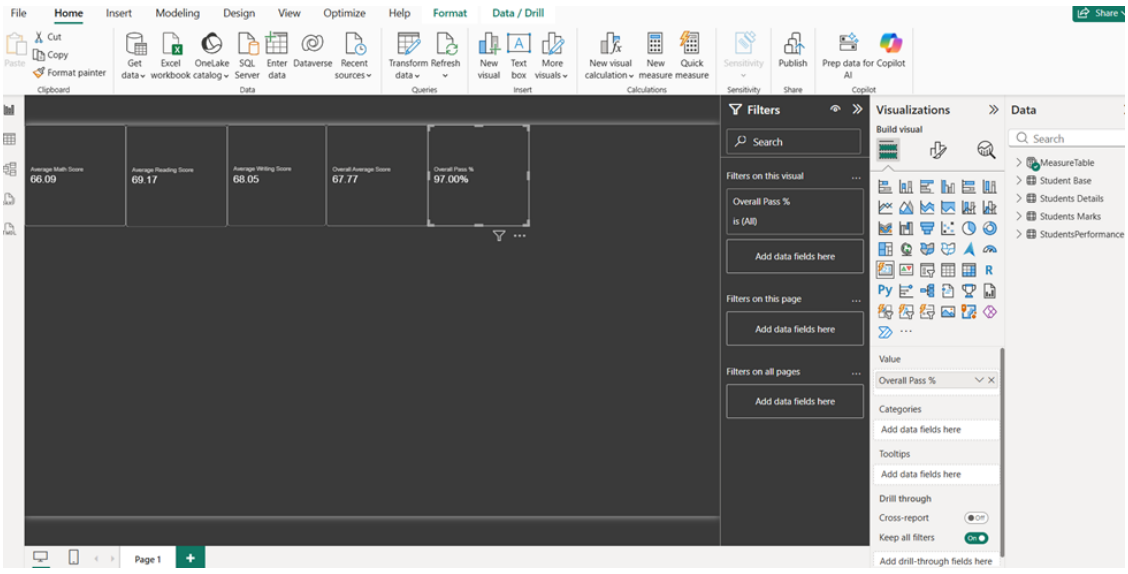
To build an interactive Student Performance dashboard in Power BI Desktop using the cleaned data model from Week 7, and to find useful insights about how students perform in Math, Reading and Writing.

Dataset

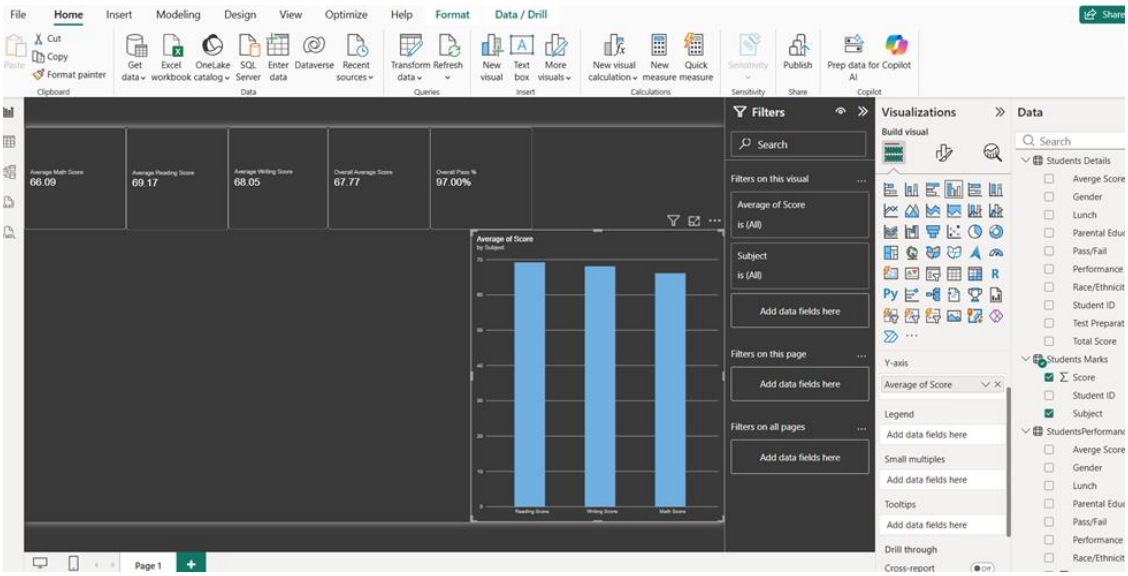
I used the Student Performance in Exams dataset that I cleaned and transformed in Week 7. The model has the tables Student Base, Students Details, Students Marks (unpivoted Subject and Score) and StudentsPerformance, along with a MeasureTable that holds my DAX measures. The columns used are Gender, Race/Ethnicity, Parental Education, Lunch, Test Preparation Course, Performance Grade, Pass/Fail and the three subject scores.

Procedure

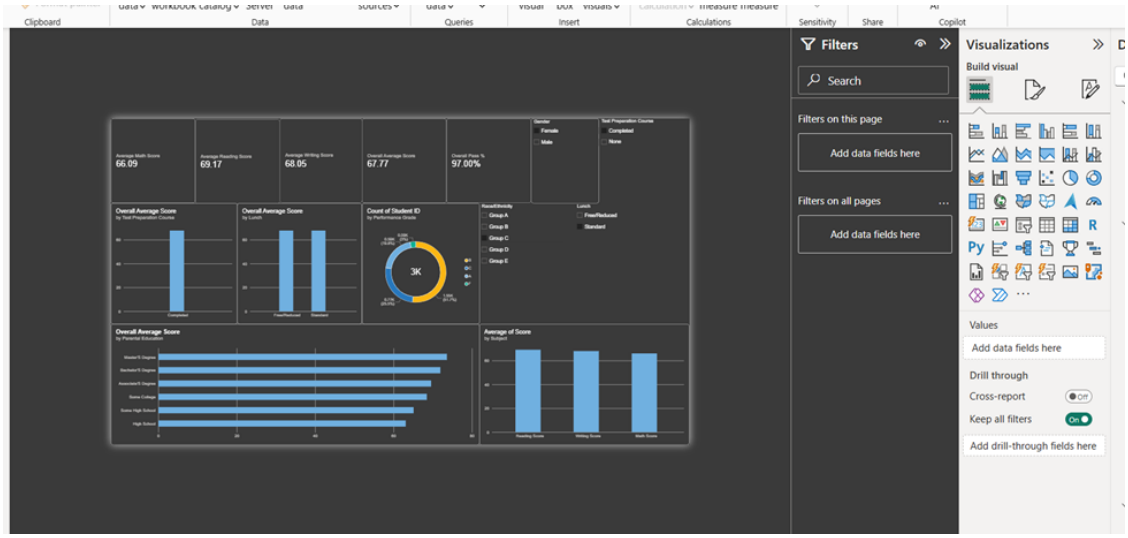
Step 1: Opened the prepared file and created KPI cards. I opened the Power BI file from Week 7 and checked the model and the measures in the MeasureTable. Then I added five card visuals for Average Math Score, Average Reading Score, Average Writing Score, Overall Average Score and Overall Pass %.



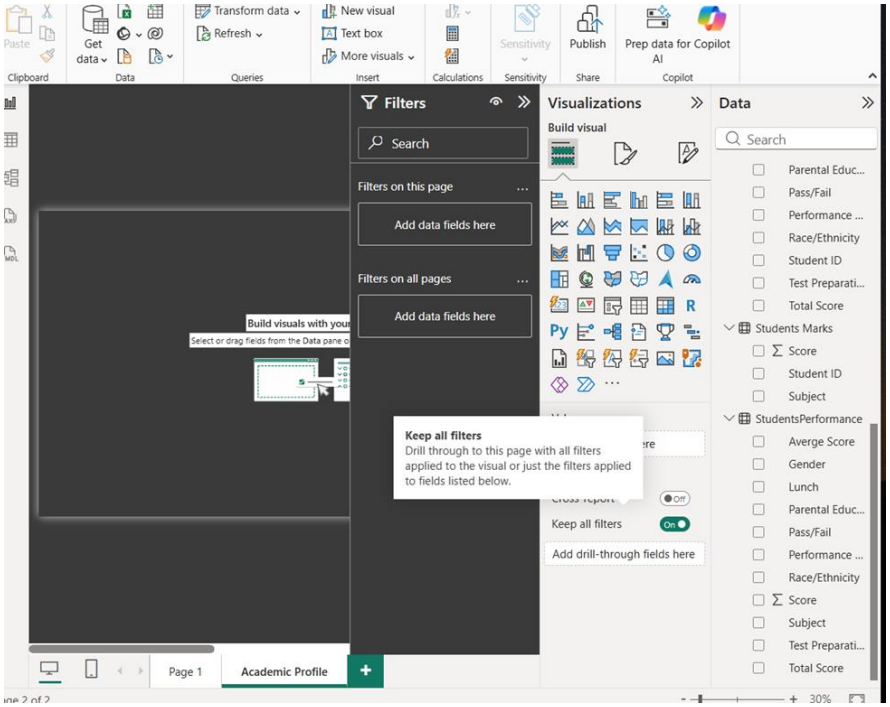
Step 2: Created the subject performance chart. I added a clustered column chart with Subject on the X-axis and the Average of Score on the Y-axis (from the Students Marks table) to compare Reading, Writing and Math.



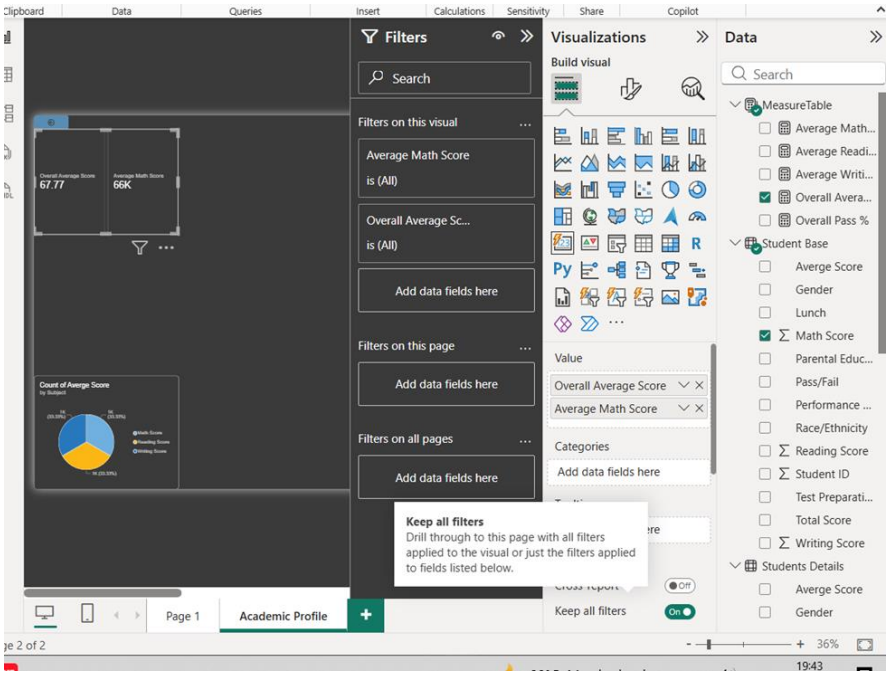
Step 3: Built the full dashboard on Page 1. I arranged all the visuals in one layout with the cards at the top. I added slicers for Gender, Test Preparation Course, Race/Ethnicity and Lunch. I also added a column chart of Overall Average Score by Test Preparation Course, a column chart by Lunch, a donut chart of the Count of Student ID by Performance Grade, a bar chart of Overall Average Score by Parental Education, and the Average of Score by Subject chart.



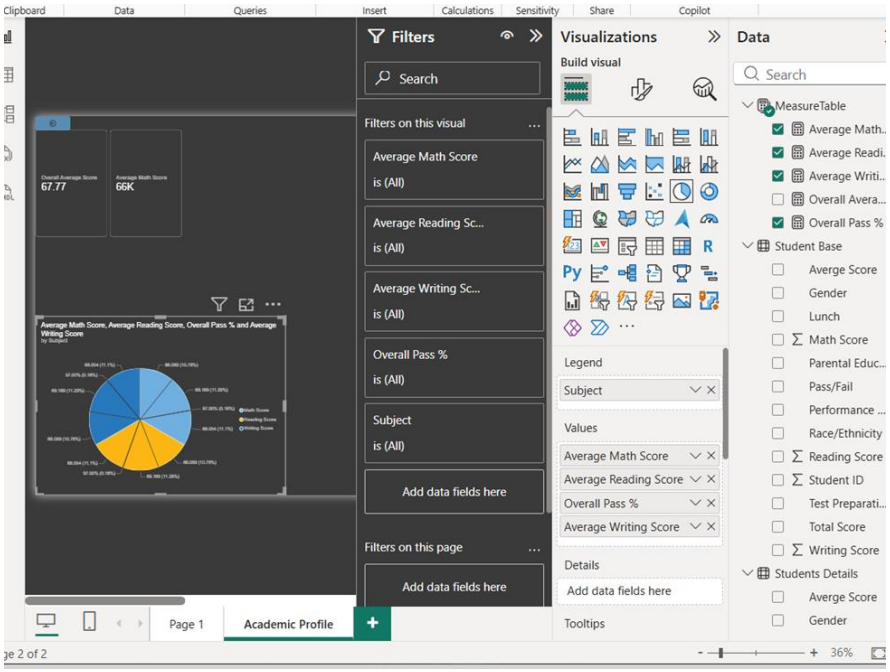
Step 4: Created the Academic Profile page. I added a new page named Academic Profile for the detailed view. In the Drill through section, the Keep all filters option was turned on.



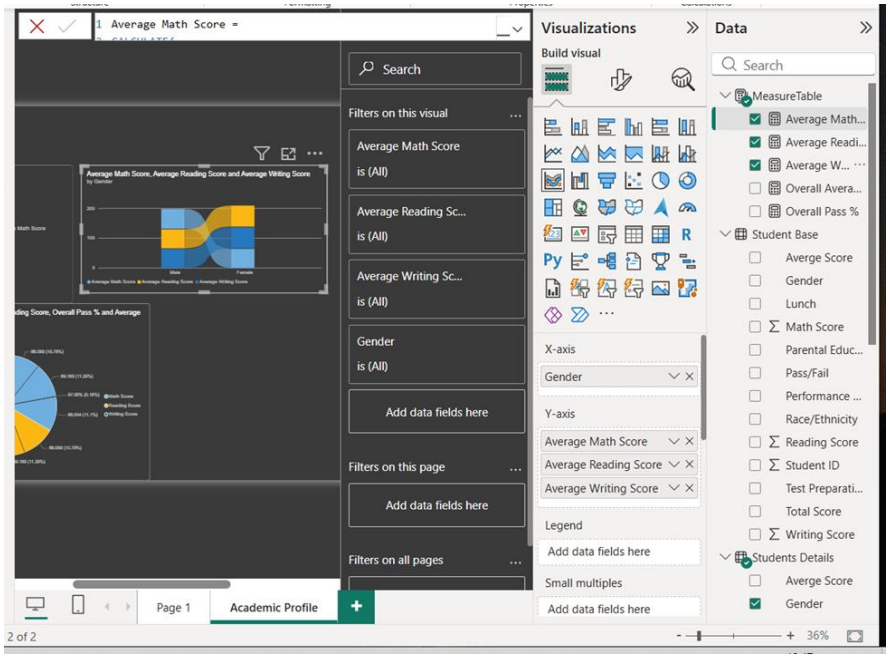
Step 5: Added detail visuals on Academic Profile. I placed cards for Overall Average Score and Average Math Score on this page. Below them I added a pie chart by Subject.



I then added the Average Math, Reading and Writing Score and Overall Pass % measures to the pie chart values with Subject as the legend.



Step 6: Added a gender comparison. I created a ribbon chart with Gender on the X-axis and the Average Math, Reading and Writing Score measures on the Y-axis to compare male and female students across the three subjects.



Step 7: Formatting. I used a dark theme for the whole report, gave the visuals clear titles, and aligned them in a grid so that the cards, charts and slicers are easy to read.

Insights

Insight 1: Overall performance is good.

- **Evidence:** The Overall Average Score is 67.77 and the Overall Pass % is 97.00%.
- **Interpretation:** Almost all the students in the dataset have passed, and the average marks are in a moderate range.

Insight 2: Reading is the strongest subject and Math is the weakest.

- **Evidence:** The average scores are Reading 69.17, Writing 68.05 and Math 66.09.
- **Interpretation:** Students are doing a little better in language subjects than in Math.

Insight 3: The three subject averages are close to each other.

- **Evidence:** The gap between the highest (Reading 69.17) and the lowest (Math 66.09) is only about 3 marks.
- **Interpretation:** Performance is fairly balanced across subjects, with no subject that is far behind.

Insight 4: Parental education is related to scores.

- **Evidence:** In the Overall Average Score by Parental Education chart, the Master's Degree group has the longest bar and the High School group has the shortest bar.
- **Interpretation:** Students whose parents have a higher education level tend to score higher in this dataset.

Insight 5: The pass rate is high but the averages leave room to improve.

- **Evidence:** The Overall Pass % is 97.00% while the Overall Average Score is only 67.77.
- **Interpretation:** Most students cross the pass mark, but many are not scoring in the higher range, so there is scope for improvement.

These insights only show relationships seen in the data. They do not prove that one factor causes another.

Validation

I checked that the measures return the expected values, that the slicers filter the visuals, and that the titles and labels are clear. I also made sure the Academic Profile page opens and that its visuals load without errors.

Conclusion

In Week 8, I built an interactive Student Performance dashboard with KPI cards, a subject comparison, parental education analysis, test preparation and lunch type analysis, a grade distribution chart, slicers and an Academic Profile page. The dashboard shows that overall performance is good, Reading is the best subject, Math is the lowest, and parental education is linked with higher scores.